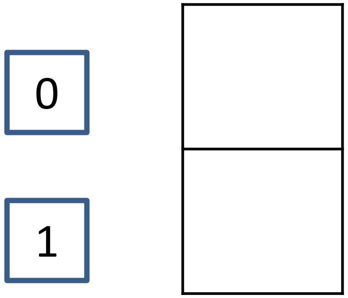
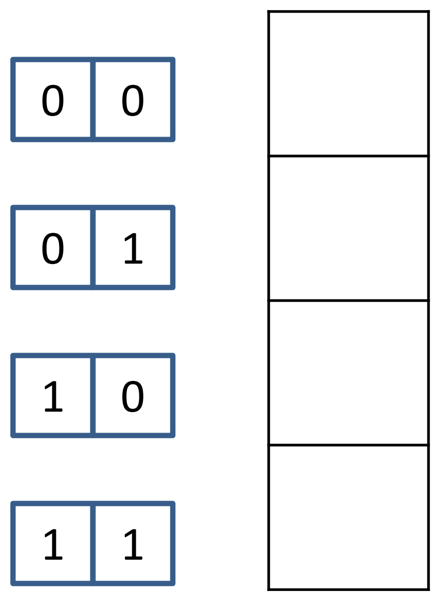
Section 11
Sheet 11 solutions

1. Memory capacity of a memory IC chip is always given in bits. Memory capacity of a computer is given in bytes.

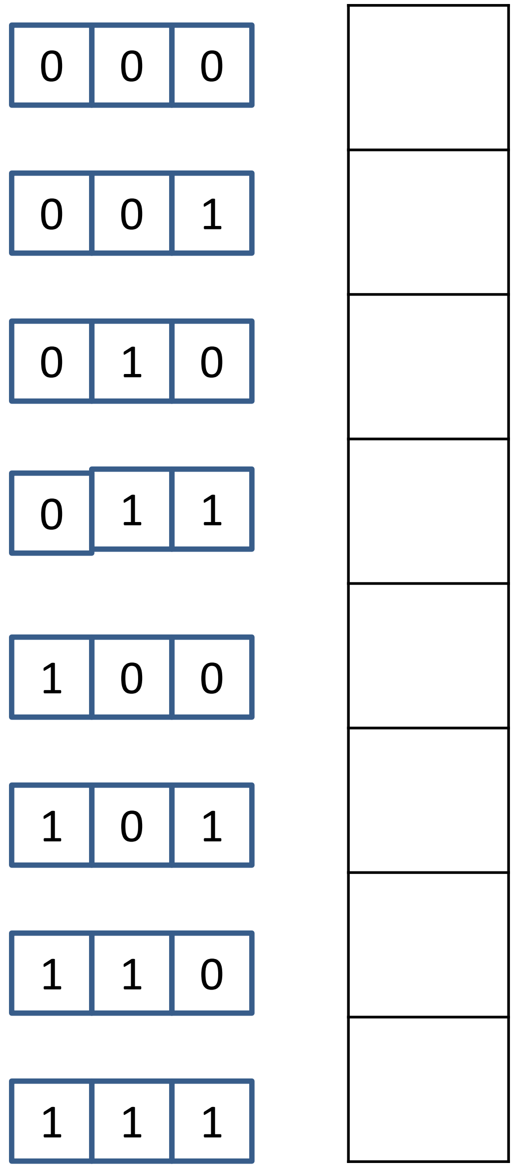
1 address pin



2 address pins



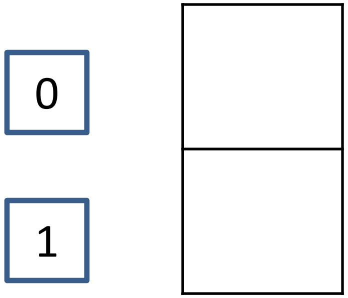
3 address pins



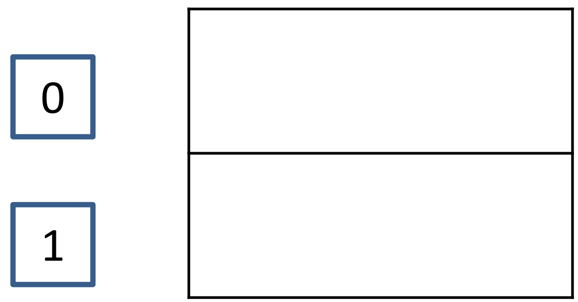
$$\text{Number of locations} = 2^{\wedge \text{number of address pins}}$$

Size of each locations = # of data pins

1 address pin

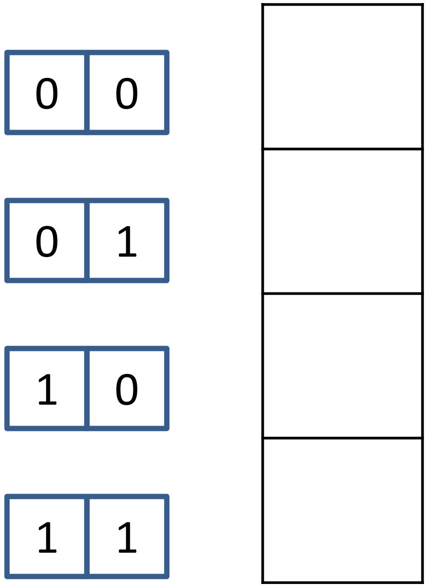


1 address pin

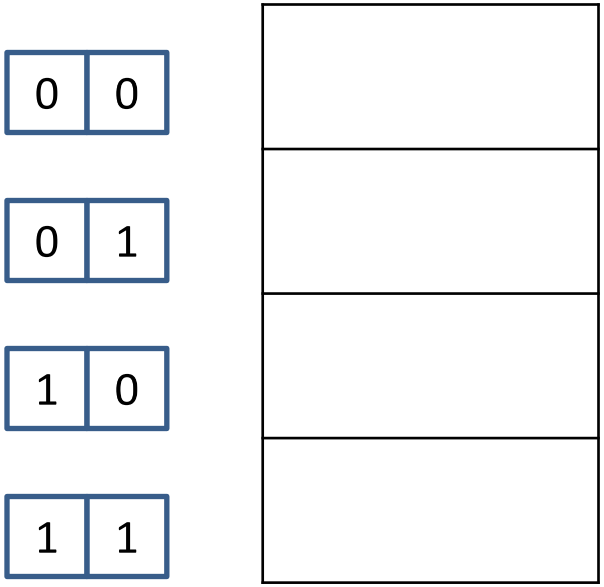


1 address pin

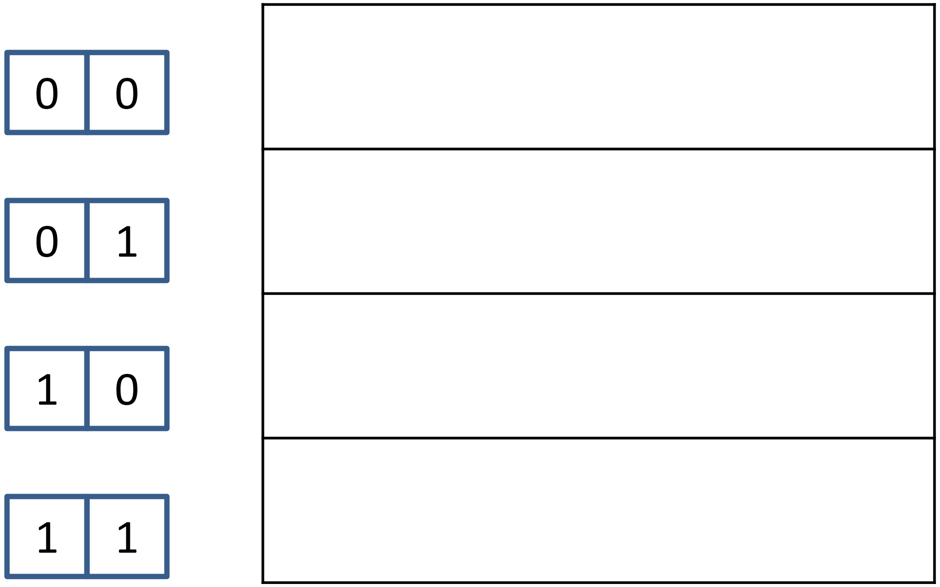
2 address pins



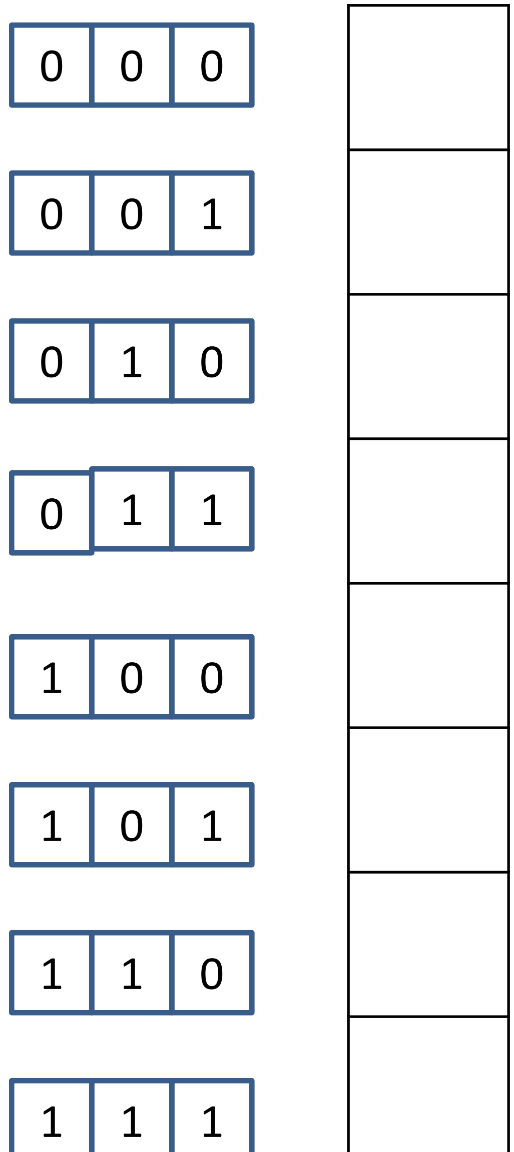
2 address pins



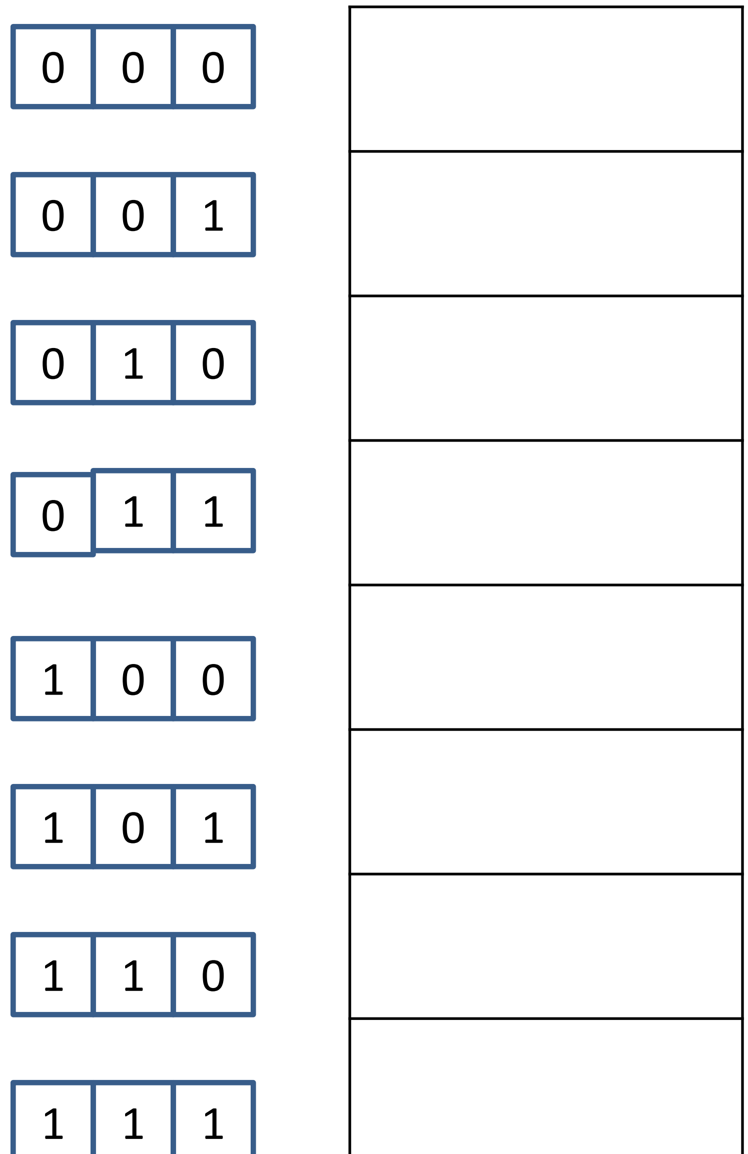
2 address pins



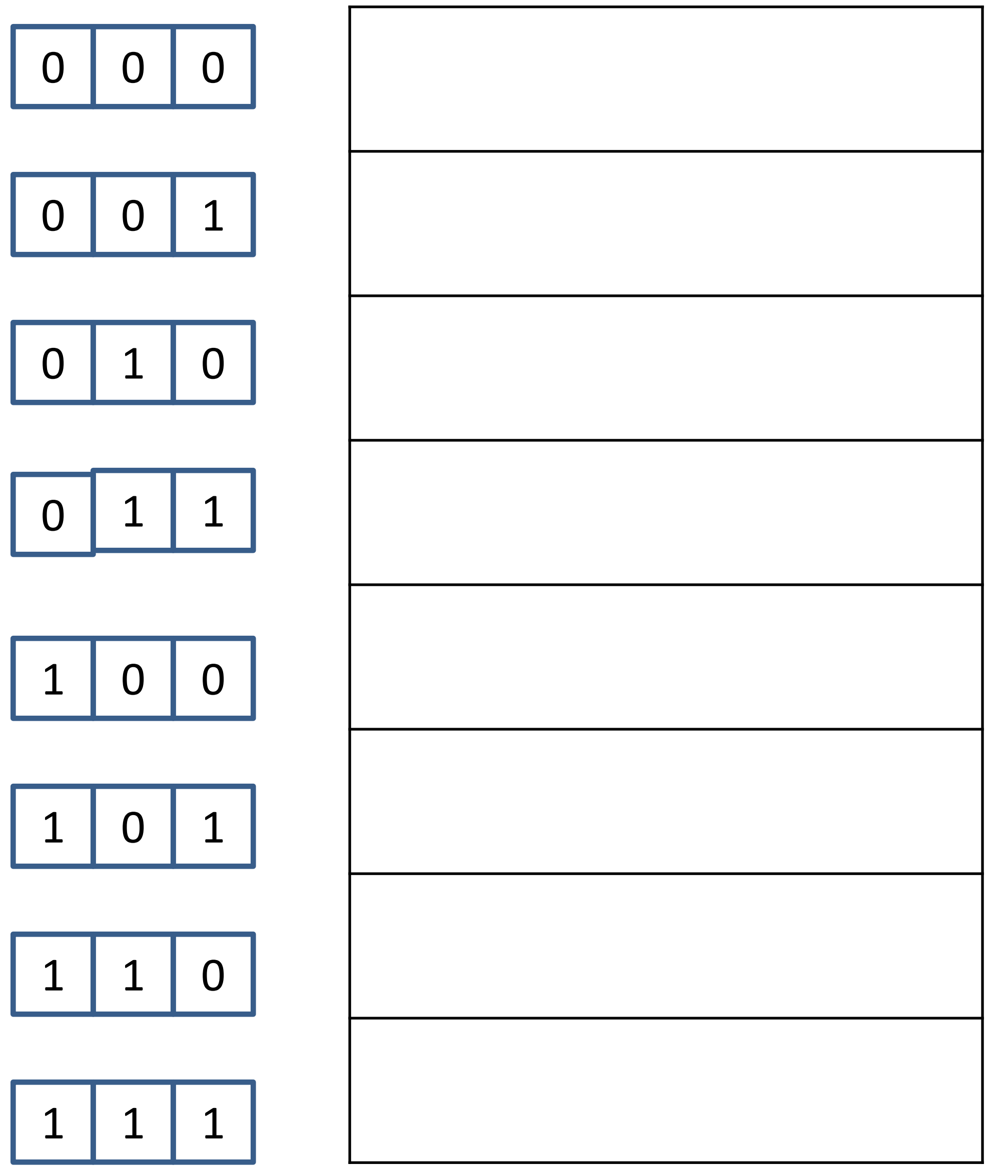
3 address pins



3 address pins



3 address pins



True or false. The more address pins, the more memory locations are inside the chip.

True or false. The more data pins, the more each location inside the chip will hold.

True or false. The more data pins, the higher the capacity of the memory chip.

True or false. With a fixed number of address pins, the more data pins, the greater the capacity of the memory chip.

2. True
3. True
4. False
5. True

6. Access time

6. Access time
7. True

6. Access time

7. True

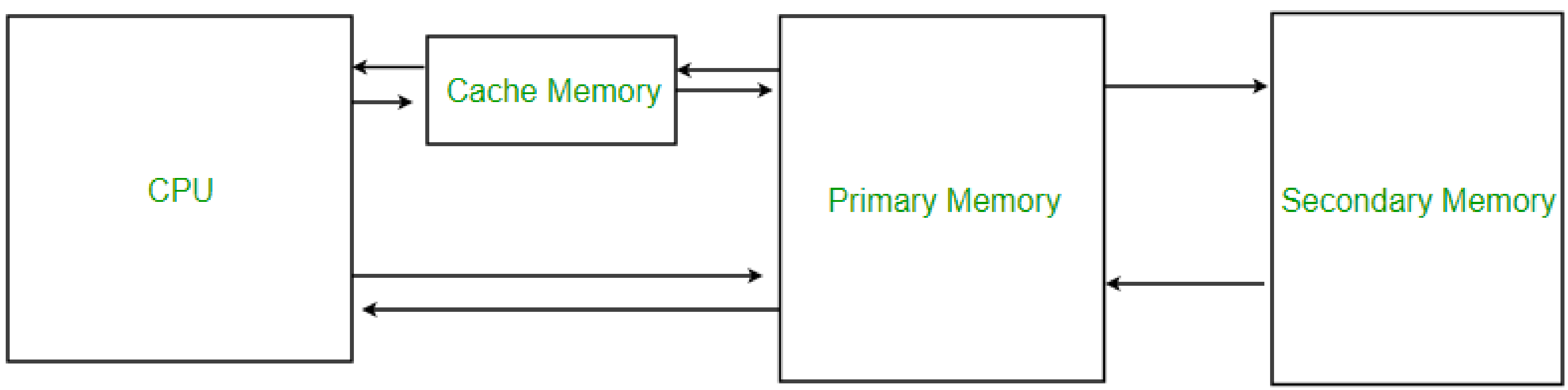
8. EEPROM can be programmed while in its system board socket, The program/erase cycle is 500,000 for Flash and EEPROM; 2000 for UV-EPROM

Brief comparison between SRAM and DRAM

Static RAM	Dynamic RAM
➢ SRAM uses transistor to store a single bit of data	➢ DRAM uses a separate capacitor to store each bit of data
➢ SRAM does not need periodic refreshment to maintain data	➢ DRAM needs periodic refreshment to maintain the charge in the capacitors for data
➢ SRAM's structure is complex than DRAM	➢ DRAM's structure is simplex than SRAM
➢ SRAM are expensive as compared to DRAM	➢ DRAM's are less expensive as compared to SRAM
➢ SRAM are faster than DRAM	➢ DRAM's are slower than SRAM
➢ SRAM are used in Cache memory	➢ DRAM are used in Main memory

9. True

9. True
10. DRAM


CPU
Cache Memory
Primary Memory
Secondary Memory

11. SRAM
12. SRAM, DRAM, cache memory

Example 10-1

A given memory chip has 12 address pins and 8 data pins. Find:
(a) the organization (b) the capacity

Solution:

(a) This memory chip has 4096 locations ($2^{12} = 4096$), and each location can hold 8 bits of data. This gives an organization of 4096×8, often represented as $4K \times 8$.

(b) The capacity is equal to 32K bits since there is a total of 4K locations and each location can hold 8 bits of data.

DRAM

Pin 4:
RAS (Row Access Strobe)
Pin 15:
CAS (Column Access Strobe)

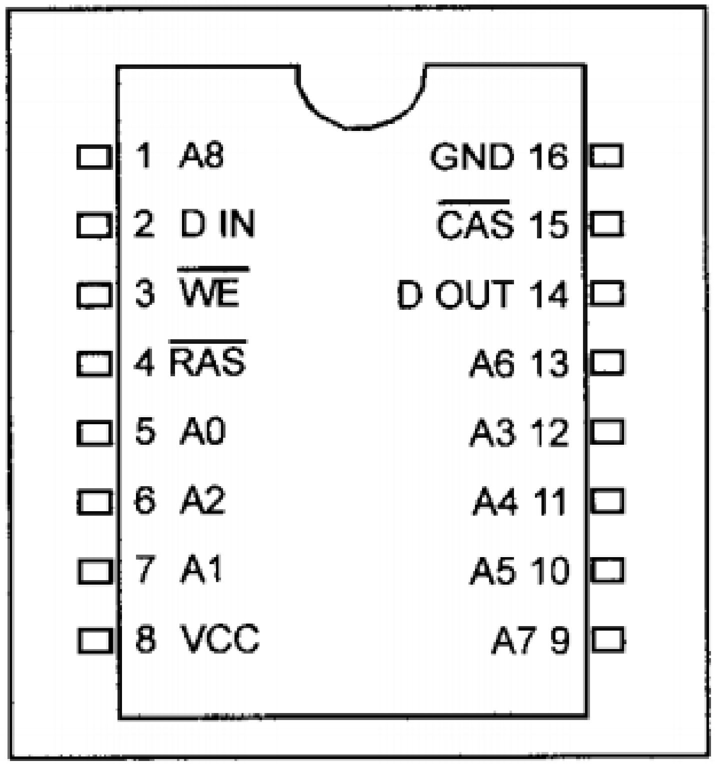


Figure 10-7. 256Kx1 DRAM

13. DRAM

14. DRAM

Example 10-5

Discuss the number of pins set aside for addresses in each of the following memory chips.

(a) 16K × 4 DRAM (b) 16K × 8 SRAM

Solution:

Since $2^{14} = 16K$:

(a) For DRAM we have 7 pins (A0–A6) for the address pins and 2 pins for RAS and CAS.

(b) For SRAM we have 14 pins (A0–A13) for address and no pins for RAS and CAS since they are associated only with DRAM.

Table 10-3: Examples of RAM Chips

Type	Part Number	Speed (ns)	Capacity	Organization	Pins
SRAM	6116-1	100	16K	2Kx8	24
	6116LP-70*	70	16K	2Kx8	24
	6264-10	100	64K	8Kx8	28
	62256LP-10*	100	256K	32Kx8	28
DRAM	4116-20	200	16K	16Kx1	16
	4116-15	150	16K	16Kx1	16
	4116-12	120	16K	16Kx1	16
	4416-12	120	64K	16Kx4	18
	4416-15	150	64K	16Kx4	18
	4164-15	150	64K	64Kx1	16
	41464-8	80	256K	64Kx4	18
	41256-15	150	256K	256Kx1	16
	41256-6	60	256K	256Kx1	16
	414256-10	100	1M	256Kx4	20
	511000P-8	80	1M	1Mx1	18
	514100-7	700	4M	4Mx1	20
NV-SRAM	DS1220	100	16K	2Kx8	24
	DS1225	150	64K	8Kx8	28
	DS1230	70	256K	32Kx8	28

* LP indicates low power.

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Find the organization and the capacity of memory chips with the following pins.

(a) EEPROM A0–A14, D0–D7

(b) UV-EPROM A0–A12, D0–D7

(c) SRAM A0–A11, D0–D7

(d) SRAM A0–A12, D0–D7

(e) DRAM A0–A10, D0

(f) SRAM A0–A12, D0-D7

(g) EEPROM A0–A11, D0–D7

(h) UV-EPROM A0–A10, D0–D7

(i) DRAM A0–A8, D0–D3

(j) DRAM A0–A7, D0–D7

Find the organization and the capacity of memory chips with the following pins.

(a) EEPROM A0–A14, D0–D7
(b) UV-EPROM A0–A12, D0–D7
(c) SRAM A0–A11, D0–D7
(d) SRAM A0–A12, D0–D7
(e) DRAM A0–A10, D0
(f) SRAM A0–A12, D0-D7
(g) EEPROM A0–A11, D0–D7
(h) UV-EPROM A0–A10, D0–D7
(i) DRAM A0–A8, D0–D3
(j) DRAM A0–A7, D0–D7

a. organization: $2^{15} \times 8 = 2^{5} \times 2^{10} \times 8 = 32K \times 8$

capacity: 256K bits

Find the organization and the capacity of memory chips with the following pins.

(a) EEPROM A0–A14, D0–D7
(b) UV-EPROM A0–A12, D0–D7
(c) SRAM A0–A11, D0–D7
(d) SRAM A0–A12, D0–D7
(e) DRAM A0–A10, D0
(f) SRAM A0–A12, D0-D7
(g) EEPROM A0–A11, D0–D7
(h) UV-EPROM A0–A10, D0–D7
(i) DRAM A0–A8, D0–D3
(j) DRAM A0–A7, D0–D7

b. organization: 2^{13} x 8 = 2^{3} x2^{10}x8 = 8Kx8

capacity: 64K bits

Find the organization and the capacity of memory chips with the following pins.

(a) EEPROM A0–A14, D0–D7
(b) UV-EPROM A0–A12, D0–D7
(c) SRAM A0–A11, D0–D7
(d) SRAM A0–A12, D0–D7
(e) DRAM A0–A10, D0
(f) SRAM A0–A12, D0-D7
(g) EEPROM A0–A11, D0–D7
(h) UV-EPROM A0–A10, D0–D7
(i) DRAM A0–A8, D0–D3
(j) DRAM A0–A7, D0–D7

c. organization: 2^{12} x 8 = 2^{2} x2^{10}x8 = 4Kx8

capacity: 32K bits

Find the organization and the capacity of memory chips with the following pins.

(a) EEPROM A0–A14, D0–D7

(b) UV-EPROM A0–A12, D0–D7

(c) SRAM A0–A11, D0–D7

(d) SRAM A0–A12, D0–D7

(e) DRAM A0–A10, D0

(f) SRAM A0–A12, D0-D7

(g) EEPROM A0–A11, D0–D7

(h) UV-EPROM A0–A10, D0–D7

(i) DRAM A0–A8, D0–D3

(j) DRAM A0–A7, D0–D7

e. organization: 2^{22} x 1 = 2^{2} x2^{20}x1 = 4Mx1

capacity: 4M bits <u>(DRAM)</u>

Find the organization and the capacity of memory chips with the following pins.

(a) EEPROM A0–A14, D0–D7
(b) UV-EPROM A0–A12, D0–D7
(c) SRAM A0–A11, D0–D7
(d) SRAM A0–A12, D0–D7
(e) DRAM A0–A10, D0
(f) SRAM A0–A12, D0-D7
(g) EEPROM A0–A11, D0–D7
(h) UV-EPROM A0–A10, D0–D7
(i) DRAM A0–A8, D0–D3
(j) DRAM A0–A7, D0–D7

i. organization: $2^{18} \times 4 = 2^{8} \times 2^{10} \times 4 = 256K \times 4$

capacity: 1024K = 1M bits (DRAM)

Q 16

Find the capacity, address, and data pins for the following memory organizations.

(a) 16K × 8 ROM

(b) 32K × 8 ROM

(c) 64K × 8 SRAM

(d) 256K × 4 DRAM

(e) 64K × 8 ROM

(f) 64K × 4 DRAM

(g) 1M × 4 DRAM

(h) 4M × 4 DRAM

(i) 64K × 8 DRAM

a. 16K x 8

capacity: 128k bits

data pins: 8 ; D0 -> D7

address pins: 16K = $2^4 x 2^{10} = 2^{14}$ // 14 pins

or address pins: $\log_2(16K)$ = 14 pins

address pins: 14; A0 -> A13

b. 32K x 8

capacity: 256k bits

data pins: 8 ; D0 -> D7

address pins: 32K = $2^5 x 2^{10} = 2^{15}$ //15 pins

or address pins: $\log_2(32K)$ = 15 pins

address pins: 15; A0 -> A14

C . 64K x 8

capacity: 512k bits

data pins: 8 ; D0 -> D7

address pins: 64K = 2^6 x2^{10} = 2^{16} //16 pins

or address pins: $\log_2(64K)$ = 16 pins

address pins: 16; A0 -> A15

d . 256K x 4 (DRAM)

capacity: 1M bits

data pins: 4 ; D0 -> D3

address pins: 256K = $2^8 x 2^{10} = 2^{18}$

or address pins: $\log_2(256K) = 18$

address pins: 9; A0 -> A8 (and 2 pins RAS and CAS)